

SQL BASIC OPERATIONS

1. CREATING DATABASE:

```
CREATE DATABASE NIT;
```

2. CREATING TABLE:

```
CREATE TABLE STUDENT (  
    NAME VARCHAR(30),  
    ID INT NOT NULL PRIMARY KEY,  
    ADDRESS VARCHAR(30),  
    MARKS INT);
```

3. DESCRIBING DATABASE:

```
DESC STUDENT;
```

OUTPUT -----

Field	Type	Null	Key	Default	Extra
NAME	varchar(30)	YES		NULL	
ID	int	NO	PRI	NULL	
ADDRESS	varchar(30)	YES		NULL	
MARKS	int	YES		NULL	

4. INSERTING DATA :

SECURE WAY -

```
INSERT INTO STUDENT (MARKS, ID, NAME, ADDRESS) VALUES (70, 12, 'PRAKASH', 'HYD');
```

INSECURE WAY -

```
INSERT INTO STUDENT VALUES ('KODI', 40, 'BANGLORE', 66);
```

INSERTING MULTIPLE VALUES -

```
INSERT INTO STUDENT VALUES ('GS', 41, 'DELHI', 88), ('MASSMAN', 60, 'GUNTUR', 78), ('MODI', 43, 'ASSAM', 96), ('AMITH', 19, 'NEW DELHI', 23);
```

5. PRINTING TABLE

1 • `SELECT * FROM STUDENT`

100% 22:1

Result Grid Filter Rows: Search

	NAME	ID	ADDRESS	MARKS
	PRAKASH	12	HYD	70
	AMITH	19	NEW DELHI	23
	KODI	40	BANGLORE	66
	GS	41	DELHI	88
	MODI	43	ASSAM	96
	MASSMAN	60	GUNTUR	78

6. SQL Arithmetic Operators

-
- 1 • **SELECT** 10+20;
 - 2 • **SELECT** 20-10;
 - 3 • **SELECT** 100*20;
 - 4 • **SELECT** 6/2
-

7. RELATIONAL OPERATORS:

- 1 • **SELECT** 10>20;
 - 2 • **SELECT** 20=20;
 - 3 • **SELECT** 20<> 23;
 - 4 • **SELECT** 50!=20;
-

8. CREATING CUSTOMERS TABLE :

USE NIT;

```
CREATE TABLE CUSTOMERS(  
    ID INT NOT NULL PRIMARY KEY,  
    NAME VARCHAR (30),  
    AGE INT,  
    SALARY FLOAT);
```

- ALTERING TABLE -----
ALTER TABLE table name
ADD column name datatype;

INSERT INTO CUSTOMERS

(ID, **NAME**, AGE, ADDRESS, SALARY)

VALUES

- (1, 'Ramesh', 32, 'Ahmedabad', 2000.00),
- (2, 'Khilan', 25, 'Delhi', 1500.00),
- (3, 'Kaushik', 23, 'Kota', 2000.00),
- (4, 'Chaitali', 25, 'Mumbai', 6500.00),
- (6, 'Komal', 22, 'MP', 4500.00),
- (7, 'Muffy', 24, 'Indore', 10000.00);

ID	NAME	AGE	SALARY	ADDRESS
1	Ramesh	32	2000	Ahmedabad
2	Khilan	25	1500	Delhi
3	Kaushik	23	2000	Kota
4	Chaitali	25	6500	Mumbai
6	Komal	22	4500	MP
7	Muffy	24	10000	Indore
NULL	NULL	NULL	NULL	NULL

9. OPERATION ON TABLE -----:

1 **SELECT * FROM CUSTOMERS WHERE SALARY > 5000;**

43:1

Result Grid

Filter Rows:

Search

Edit:

Exp

ID	NAME	AGE	SALARY	ADDRESS	
4	Chaitali	25	6500	Mumbai	
7	Muffy	24	10000	Indore	
NULL	NULL	NULL	NULL	NULL	

2 • **SELECT** * **FROM** CUSTOMERS **WHERE** SALARY =2000;

44:2

Result Grid

Filter Rows:

Search

Edit:

Ex

ID	NAME	AGE	SALARY	ADDRESS	
1	Ramesh	32	2000	Ahmedabad	
3	Kaushik	23	2000	Kota	
NULL	NULL	NULL	NULL	NULL	

3 • **SELECT** * **FROM** CUSTOMERS **WHERE** SALARY !=2000;

39:3

Result Grid

Filter Rows:

Search

Edit:

Ex

ID	NAME	AGE	SALARY	ADDRESS	
2	Khilan	25	1500	Delhi	
4	Chaitali	25	6500	Mumbai	
6	Komal	22	4500	MP	
7	Muffy	24	10000	Indore	
NULL	NULL	NULL	NULL	NULL	

4 • **SELECT** * **FROM** CUSTOMERS **WHERE** SALARY <> 2000;

41:4

Result Grid

Filter Rows:

Search

Edit:

Exp

ID	NAME	AGE	SALARY	ADDRESS	
2	Khilan	25	1500	Delhi	
4	Chaitali	25	6500	Mumbai	
6	Komal	22	4500	MP	
7	Muffy	24	10000	Indore	
NULL	NULL	NULL	NULL	NULL	

5 • **SELECT** * **FROM** CUSTOMERS **WHERE** SALARY >= 6500;

46:5

Result Grid

Filter Rows:

Q

Search

Edit:

Export

ID	NAME	AGE	SALARY	ADDRESS	
4	Chaitali	25	6500	Mumbai	
7	Muffy	24	10000	Indore	
NULL	NULL	NULL	NULL	NULL	

6 • **SELECT * FROM CUSTOMERS WHERE AGE >= 25 AND SALARY >= 6500;**

57:6		Result Grid		Filter Rows: Search	Edit:	Export/Import:
ID	NAME	AGE	SALARY	ADDRESS		
4	Chaitali	25	6500	Mumbai		
NULL	NULL	NULL	NULL	NULL		

7 • **SELECT * FROM CUSTOMERS WHERE AGE >= 25 OR SALARY >= 6500;**

41:7		Result Grid		Filter Rows: Search	Edit:	Export/Import:
ID	NAME	AGE	SALARY	ADDRESS		
1	Ramesh	32	2000	Ahmedabad		
2	Khilan	25	1500	Delhi		
4	Chaitali	25	6500	Mumbai		
7	Muffy	24	10000	Indore		
NULL	NULL	NULL	NULL	NULL		

8 • **SELECT * FROM CUSTOMERS WHERE AGE IS NOT NULL;**

46:8		Result Grid		Filter Rows: Search	Edit:	Export/Import:
ID	NAME	AGE	SALARY	ADDRESS		
1	Ramesh	32	2000	Ahmedabad		
2	Khilan	25	1500	Delhi		
3	Kaushik	23	2000	Kota		
4	Chaitali	25	6500	Mumbai		
6	Komal	22	4500	MP		
7	Muffy	24	10000	Indore		
NULL	NULL	NULL	NULL	NULL		

9 • **SELECT * FROM CUSTOMERS WHERE NAME LIKE 'Ko%';**

46:9		Result Grid		Filter Rows: Search	Edit:	Export/Import:
ID	NAME	AGE	SALARY	ADDRESS		
6	Komal	22	4500	MP		
NULL	NULL	NULL	NULL	NULL		

10 • **SELECT * FROM CUSTOMERS WHERE AGE IN (25, 27);**

45:10		Result Grid		Filter Rows: Search	Edit:	Export/Import:
ID	NAME	AGE	SALARY	ADDRESS		
2	Khilan	25	1500	Delhi		
4	Chaitali	25	6500	Mumbai		
NULL	NULL	NULL	NULL	NULL		

11 • **SELECT** * **FROM** CUSTOMERS **WHERE** AGE **BETWEEN** 25 **AND** 27;

52:11

Result Grid Filter Rows: Search Edit: Export/Import:

ID	NAME	AGE	SALARY	ADDRESS
2	Khilan	25	1500	Delhi
4	Chaitali	25	6500	Mumbai
NULL	NULL	NULL	NULL	NULL

12 • **SELECT** AGE **FROM** CUSTOMERS **WHERE EXISTS** (**SELECT** AGE **FROM** CUSTOMERS **WHERE** SALARY>6500);

72:12

Result Grid Filter Rows: Search Export:

AGE
32
25
23
25
22
24

13 • **SELECT** * **FROM** CUSTOMERS **WHERE** AGE > **ALL** (**SELECT** AGE **FROM** CUSTOMERS **WHERE** SALARY>6500);

87:13

Result Grid Filter Rows: Search Edit: Export/Import:

ID	NAME	AGE	SALARY	ADDRESS
1	Ramesh	32	2000	Ahmedabad
2	Khilan	25	1500	Delhi
4	Chaitali	25	6500	Mumbai
NULL	NULL	NULL	NULL	NULL

10. SQL - Boolean Expressions:

1 • **SELECT** * **FROM** CUSTOMERS **WHERE** SALARY = 10000;

45:1

Result Grid Filter Rows: Search Edit: Exp

ID	NAME	AGE	SALARY	ADDRESS
7	Muffy	24	10000	Indore
NULL	NULL	NULL	NULL	NULL

11. SQL NUMERICAL EXPRESSIONS :

1 • **SELECT** (15+6) **AS** ADDITION

26:1

Result Grid Filter Rows: Search

AD...
21

```
1 • SELECT COUNT(*) AS "RECORDS" FROM CUSTOMERS;
```

- | |
|---|
| 12:1 |
| Result Grid   Filter Rows: <input type="text" value="Search"/> Export:  |
| RECORDS |
| 6 |

12. DATE EXPRESSIONS :

```
1 • SELECT current_timestamp();
```

- | |
|--|
| 28:1 |
| Result Grid   Filter Rows: <input type="text" value="Search"/> Ex |
| current_timestam... |
| 2026-06-09 16:42:01 |

13. CREATING / DROP / DELETE DATABASES :

- 1 • **CREATE DATABASE** DUMMYDB;
- 2 • **SHOW DATABASES;**

•

↕	24:1	
Result Grid  Filter Rows: Search		
	Database	
	DUMMY	
☒	DUMMYDB	

- 1 • **DROP DATABASE** DUMMY;
- 2 • **DROP DATABASE** DUMMYDB;
- 3 • **SHOW DATABASES;**

•

↕	16:3	
Result Grid  Filter Rows: Search		
	Database	
	information_schema	
☒	mysql	
	NIT	
☒	performance_schema	
	sys	

14. SELECT SQL QUERY :

1 **SELECT** ID, **NAME**, SALARY **FROM** CUSTOMERS;

9:1

Result Grid

Filter Rows:

Search

Edit:

ID	NAME	SALARY	
1	Ramesh	2000	
2	Khilan	1500	
3	Kaushik	2000	
4	Chaitali	6500	
6	Komal	4500	
7	Muffy	10000	
NULL	NULL	NULL	